

Large-Scale and Multi-Structured Databases

JAVA Recap

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Objective of this Module

Help students recall and refresh their Java programming skills.

PREREQUISITE

We assume that students are already able to work with Java.

REVIEW

We will review some basic elements of Java.

PRACTICE

Students will solve some programming exercises using an IDE (Eclipse or IntelliJ IDEA).

The Java Buzzwords

CONCEPT

Simple

Secure

Portable

Object-Oriented

Robust

Multithreaded

**Architecture-
neutral**

Interpreted

**High
performance**

Distributed

... and more

JAVA BASICS

CLASSES & OBJECTS

POLYMORPHISM

TYPES & METHODS

EXERCISES

Some Salient Characteristics of Java

CONCEPT

PLATFORM INDEPENDENT

The same program can run on any correctly implemented Java system

OBJECT-ORIENTED

Structured in terms of **classes**, which group data with operations on that data
Can construct new classes by **extending** existing ones

DESIGNED AS

A **core language** plus a rich collection of **commonly available packages**

WIDELY USED

For server-side and data-intensive applications

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EXERCISES

Program Structure

CONCEPT

Typical Java program consists of:

- User written classes
- Java **API** classes (Application Programming Interface)

Java application:

- Has one class with a **main** method

Java program basic elements:

Packages

Classes

Data fields

Methods

JAVA BASICS

CLASSES & OBJECTS

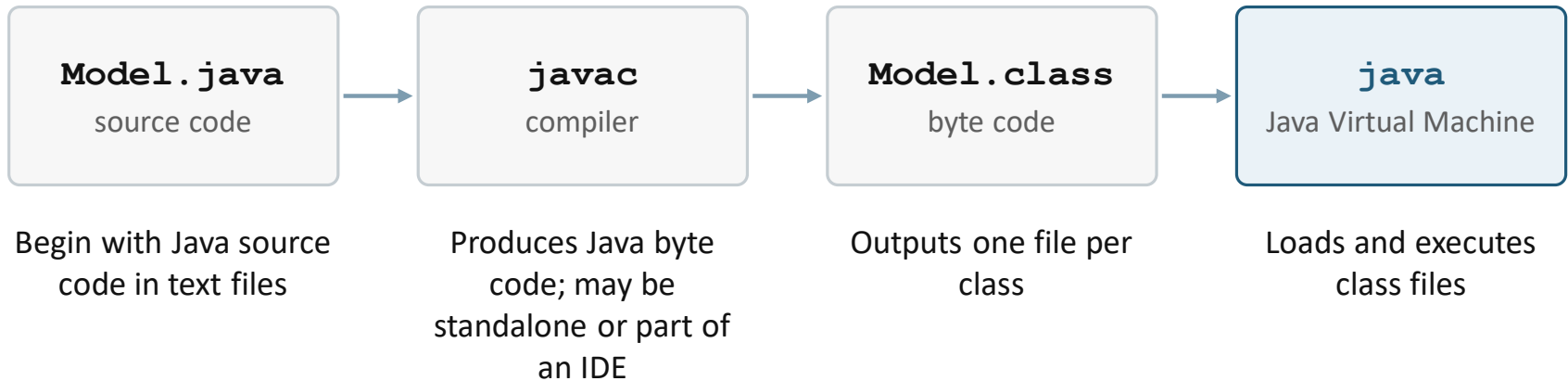
POLYMORPHISM

TYPES & METHODS

EXERCISES

Java Processing and Execution

CONCEPT



The JVM may compile class files to native code (e.g., x86) internally.

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EXERCISES

Classes and Objects

CONCEPT

The class is the unit of programming

A JAVA PROGRAM	A CLASS	CHARACTERISTICS
A collection of classes Each class definition (usually) in its own .java file	Describes objects (instances) and their common characteristics All the instances have these same characteristics	Data fields for each object Methods (operations) that do work on the objects

The file name must match the class name!

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EXERCISES

Classes and Objects

EXAMPLE

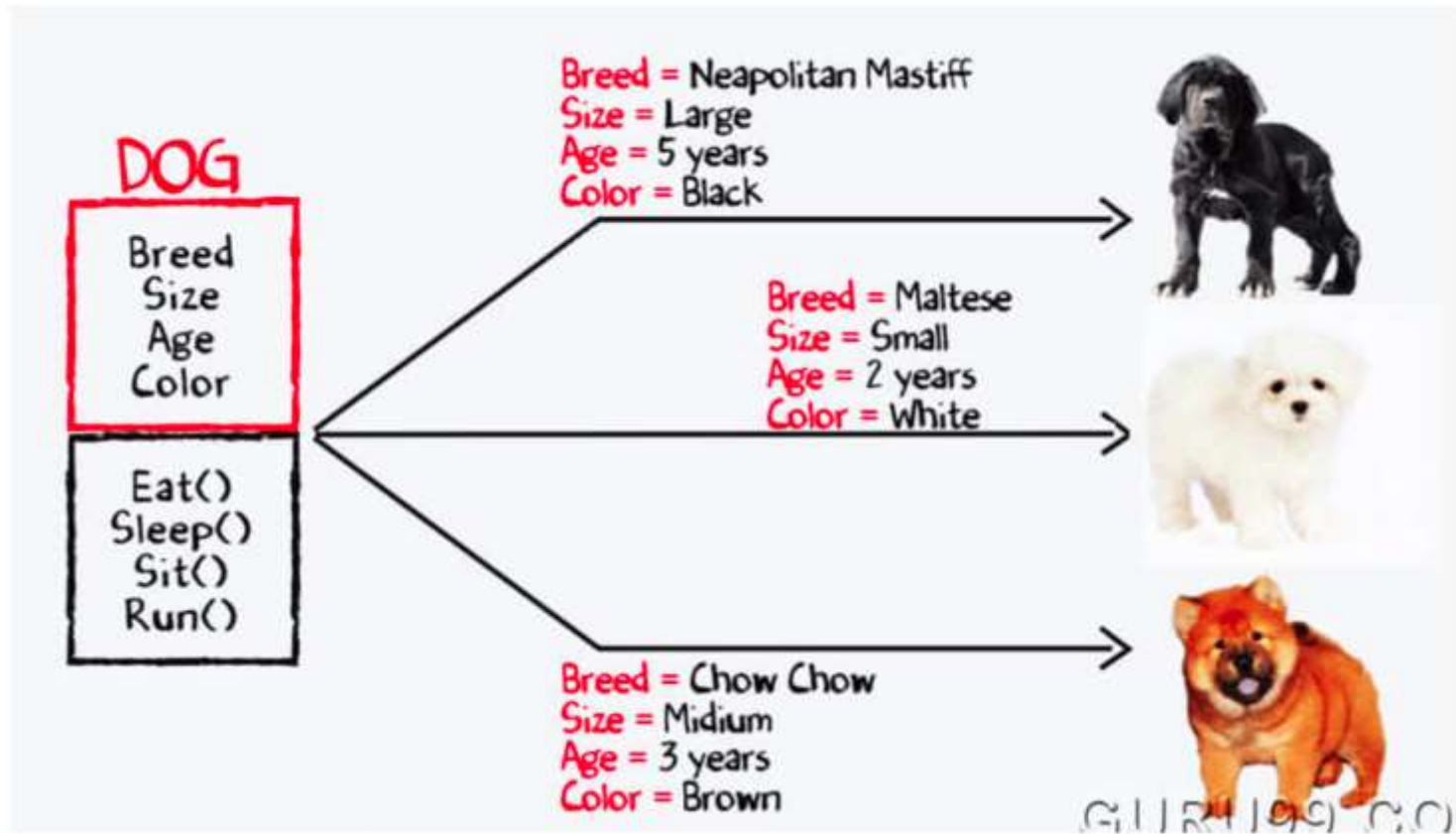


Image extracted from: <https://www.guru99.com/java-oops-class-objects.html>

Classes and Objects

EXAMPLE

```
// Class Declaration
public class Dog {
    // Instance Variables
    String breed;
    String size;
    int age;
    String color;

    // method 1
    public String getInfo() {
        return ("Breed is: "+breed+" Size is:"+size+" Age is:"+age+" color is: "+color);
    }

    public static void main(String[] args) {
        Dog maltese = new Dog();
        maltese.breed="Maltese";
        maltese.size="Small";
        maltese.age=2;
        maltese.color="white";
        System.out.println(maltese.getInfo());
    }
}
```

Image extracted from: <https://www.guru99.com/java-oops-class-objects.html>

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Static Classes

EXAMPLE

```
public class OuterClass {  
    // Outer class static variable  
    static String outerStaticVariable = "Outer Static Variable";  
  
    // Static nested class  
    static class NestedStaticClass {  
        // Method in the static nested class  
        public void displayMessage() {  
            System.out.println("This is a static nested class.");  
            System.out.println("Accessing outer class static variable: " + outerStaticVariable);  
        }  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        // Creating an instance of the static nested class  
        OuterClass.NestedStaticClass nested = new OuterClass.NestedStaticClass();  
        nested.displayMessage();  
    }  
}
```

Polymorphism

CONCEPT

Polymorphism is the ability to create a variable, a function, or an object that has more than one form.

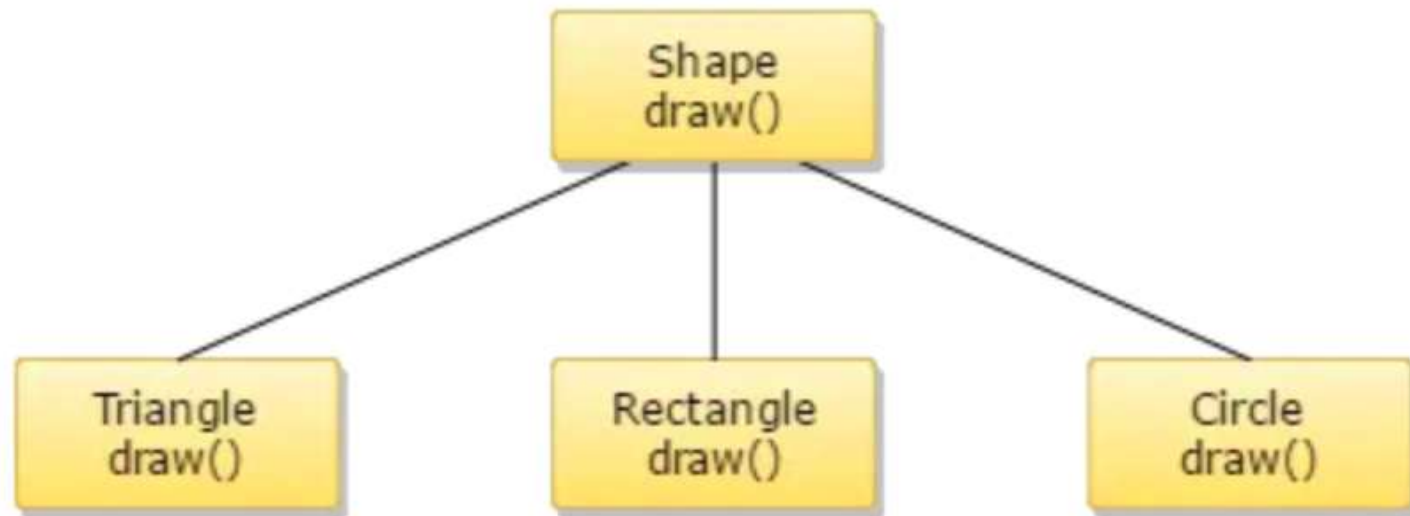


Image extracted from: <https://www.w3schools.in/java/polymorphism/>

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Compile time Polymorphism

EXAMPLE

```
public class Calculator {  
  
    public Integer sum(Integer a, Integer b) {  
        return a + b;  
    }  
  
    public Float sum(Float a, Float b) {  
        return a + b;  
    }  
  
    public Double sum(Double a, Double b) {  
        return a + b;  
    }  
}
```

```
Calculator calc = new Calculator();  
  
Integer sum1 = calc.sum(1, 2);  
Float sum2 = calc.sum(1f, 2f);  
Double sum3 = calc.sum(1d, 2d);
```

Image extracted from: <https://howtodoinjava.com/java/oops/what-is-polymorphism-in-java/>

Run time Polymorphism

EXAMPLE

```
public class Animal {
    public void makeNoise()
    {
        System.out.println("Some sound");
    }
}

class Dog extends Animal{
    public void makeNoise()
    {
        System.out.println("Bark");
    }
}

class Cat extends Animal{
    public void makeNoise()
    {
        System.out.println("Meawoo");
    }
}
```

```
public class Demo
{
    public static void main(String[] args) {
        Animal a1 = new Cat();
        a1.makeNoise(); //Prints Meawoo

        Animal a2 = new Dog();
        a2.makeNoise(); //Prints Bark
    }
}
```

Image extracted from: <https://howtodoinjava.com/java/oops/what-is-polymorphism-in-java/>

Grouping Classes: The Java API

CONCEPT

API = *Application Programming Interface*

Java = small core + extensive collection of packages

PACKAGE	IMPORT	MAIN
A package consists of some related Java classes	The import statement tells the compiler to make available classes and methods of another package	A main method indicates where to begin executing a class (if it is designed to be run as a program)

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EXERCISES

References and Primitive Data Types

CONCEPT

Java distinguishes two kinds of entities

PRIMITIVE TYPES

Primitive-type data is stored in
primitive-type variables

OBJECTS

Reference variables store the
address of an object

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EXERCISES

Primitive Data Types

CONCEPT

Represent numbers, characters, boolean values

INTEGERS

`byte`
`short`
`int`
`long`

REAL NUMBERS

`float`
`double`

CHARACTERS

`char`

BOOLEAN

`true`
`false`

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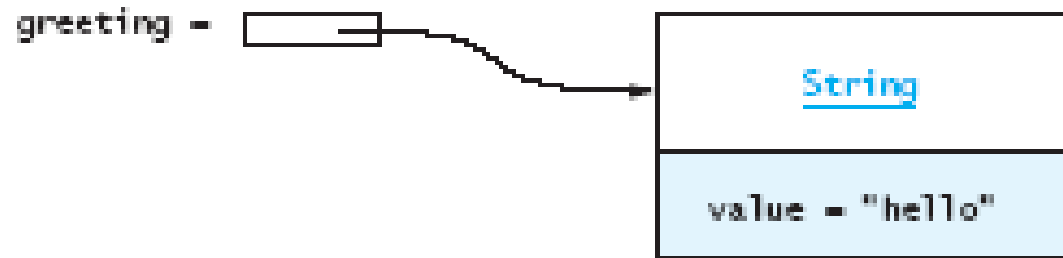
EXERCISES

Referencing and Creating Objects

CONCEPT

- We can **declare reference variables**, which reference objects of **specified types**
- Two reference variables can reference **the same object**
- The **new** operator creates an instance of a class
- A **constructor** executes when a new object is created

Example: `String greeting = "hello";`



Methods

CONCEPT

A Java method defines a group of statements as performing a particular operation

STATIC

static indicates a *static* or *class* method

INSTANCE

A method that is not **static** is an *instance* method

All method arguments are *call-by-value*

PRIMITIVE TYPE

The **value** is passed to the method
The method may modify its local copy **but**
will not affect the caller's value

OBJECT REFERENCE

The **address of the object** is passed
A change to the reference variable does not
affect the caller
But operations can affect the object, visible
to the caller

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EXERCISES

Exercise 0

CHALLENGE

- 1 Install a recent JDK (e.g., 21 or 25 LTS) and your preferred IDE (with Maven support)

Eclipse	IntelliJ IDEA	NetBeans
https://www.eclipse.org/	https://www.jetbrains.com/idea/	https://netbeans.apache.org/

- 2 Create a Java Project
- 3 Write and run your first “Hello World!” program.

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EXERCISES

Exercise 0

CHALLENGE

How long does it take?

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EXERCISES

Exercise 1

CHALLENGE

Write a Java program that takes three numbers from the user and prints the greatest number.

INPUT DATA

```
Input the 1st number: 25  
Input the 2nd number: 78  
Input the 3rd number: 87
```

EXPECTED OUTPUT

```
The greatest: 87
```

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EXERCISES

Exercise 2

CHALLENGE

Write a Java method to count all lowercase vowels in a string.

INPUT DATA

Input the string: unipi

EXPECTED OUTPUT

Number of Vowels in the
string: 3

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Exercise 3

CHALLENGE

Write a Java program to retrieve the elements at specified indexes from a given ArrayList.

TEST DATA

Create the ArrayList [Red, Green, Orange, White, Black] and retrieve the first and the third element

EXPECTED OUTPUT

First element: Red
Third element: Orange

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EXERCISES

Exercise 4 (homework)

HOMEWORK

Write a Java program that reads N numbers from the user and prints the greatest one.

The first line specifies how many numbers (N) to read; the second line contains the N numbers separated by spaces.

INPUT

```
6
25 78 87 154 -45 2023
```

OUTPUT

```
2023
```

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EXERCISES

Exercise 5 (homework)

HOMEWORK

Given a list L of $N - 1$ numbers, find the missing number.

The first line specifies the value of N .

CONSTRAINTS

$1 \leq L[i] \leq N$
 $0 \leq i < N - 1$
No duplicate numbers
allowed

INPUT

6
1 5 6 3 2

OUTPUT

4

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EXERCISES

Suggested Readings

Mitsunori Ogihara

Fundamentals of Java Programming

Springer, 2018

Online resources

<https://docs.oracle.com/javase/tutorial/>

Check books on onerearch.unipi.it